DATA201 Data Wrangling Pipeline Report

Group 7

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Overview

This report summarises the outputs from the automated DATA201 data wrangling pipeline.

The pipeline processes the Airbnb and tenancy datasets through Deliverables 3, 4 and 5. The Makefile controls the order of each stage and automatically rebuilds outputs when required.

Deliverable 3 — Data Preparation and Initial Analysis

Combined Airbnb Data

```
combined_listings <- read_csv(
  "output_data/Deliverable3_output/combined_listings.csv",
  show_col_types = FALSE
)

glimpse(combined_listings)
```

```
## Rows: 35,796
## Columns: 20
## $ id                                <dbl> 4.018377e+07, 1.149729e+18, 1.097487e+0~
## $ name                              <chr> "Ensuite Cabin (Sleeps 6)", "Handy city~
## $ host_id                           <dbl> 98758734, 298456605, 4993285, 54299306,~
## $ host_name                         <chr> "North South Holiday Park", "Tony", "Pa~
## $ neighbourhood_group               <chr> "Christchurch City", "Christchurch City~
## $ neighbourhood                     <chr> "Harewood Ward", "Central Ward", "Fenda~
## $ latitude                          <dbl> -43.47002, -43.53270, -43.51149, -43.48~
## $ longitude                         <dbl> 172.5709, 172.6640, 172.6254, 172.6687,~
## $ room_type                         <chr> "Private room", "Private room", "Privat~
## $ price                             <dbl> NA, NA, 166, NA, 126, 585, 145, 450, NA~
## $ minimum_nights                    <dbl> 1, 1, 1, 2, 1, 2, 1, 2, 1, 1, 7, 1, 1, ~
## $ number_of_reviews                 <dbl> 0, 25, 230, 2, 8, 0, 6, 0, 1, 0, 1, 5, ~
## $ last_review                       <date> NA, 2025-07-15, 2025-03-03, 2025-04-26~
## $ reviews_per_month                 <dbl> NA, 1.47, 2.17, 0.35, 0.58, NA, 0.26, N~
## $ calculated_host_listings_count    <dbl> 20, 6, 1, 1, 20, 3, 4, 1, 2, 6, 4, 14, ~
## $ availability_365                  <dbl> 317, 0, 221, 0, 359, 78, 348, 363, 4, 0~
## $ number_of_reviews_ltm             <dbl> 0, 21, 14, 2, 5, 0, 2, 0, 0, 0, 1, 4, 2~
## $ license                           <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,~
## $ month                             <chr> "October", "October", "October", "Octob~
## $ year                              <dbl> 2025, 2025, 2025, 2025, 2025, 2025, 202~
```

Example Data

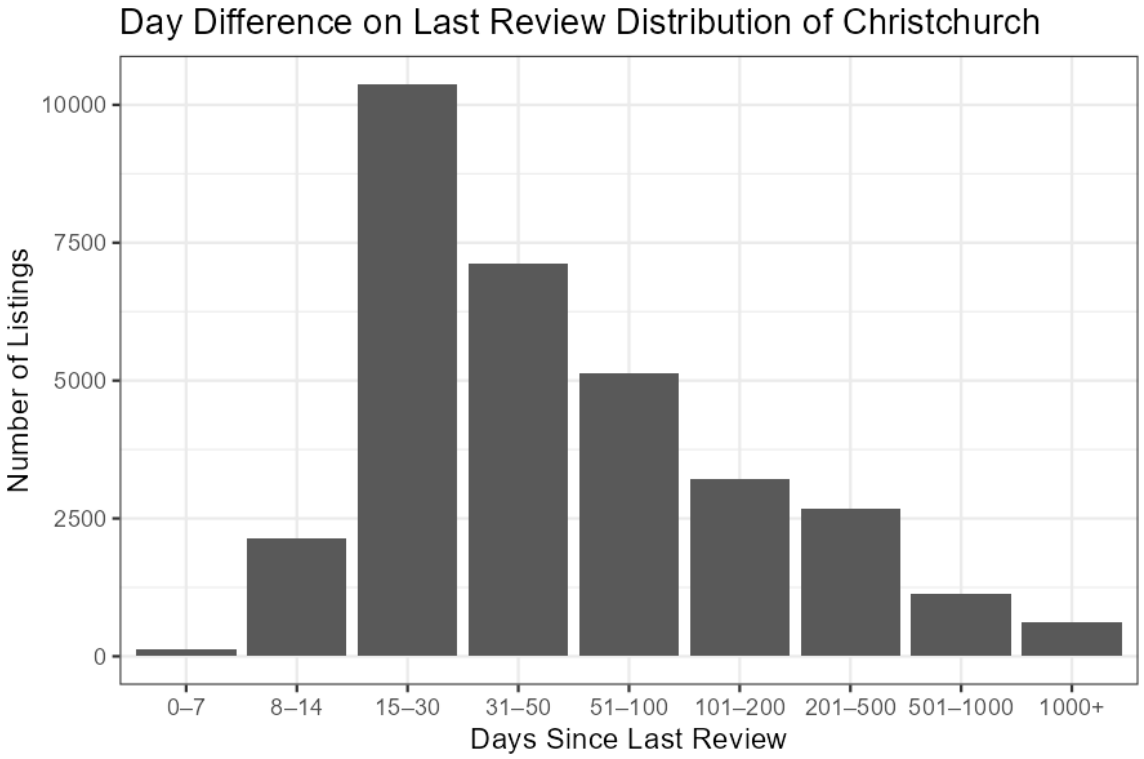
Only the most useful columns are shown below to keep the table readable.

```
combined_listings |>
  select(
    id,
    neighbourhood,
    room_type,
    price,
    number_of_reviews,
    month,
    year
  ) |>
  head(6) |>
  kable(
    caption = "Example Airbnb Listings",
    digits = 2
  )
```

Table 1: Example Airbnb Listings

id	neighbourhood	room_type	price	number_of_reviews	month	year
4.018377e+07	Harewood Ward	Private room	NA	0	October	2025
1.149729e+18	Central Ward	Private room	NA	25	October	2025
1.097487e+06	Fendalton Ward	Private room	166	230	October	2025
1.339648e+18	Burwood Ward	Entire home/apt	NA	2	October	2025
1.211070e+18	Riccarton Ward	Private room	126	8	October	2025
1.454661e+18	Fendalton Ward	Entire home/apt	585	0	October	2025

Date Difference



Price Distribution



Summary Statistics

```
summary_statistics <- read_csv(
  "output_data/Deliverable3_output/summary_statistics.csv",
  show_col_types = FALSE
)

summary_statistics |>
  select(
    variable,
    type,
    total_rows,
    missing,
    missing_percent,
    unique_values
  ) |>
  kable(
    caption = "Dataset Summary Statistics",
    digits = 2
  )
```

Table 2: Dataset Summary Statistics

variable	type	total_rows	missing	missing_percent	unique_values
id	identifier	35796	0	0.00	4264
name	categorical/text	35796	0	0.00	5170
host_id	identifier	35796	0	0.00	2171

variable	type	total_rows	missing	missing_percent	unique_values
host_name	categorical/text	35796	1	0.00	1460
neighbourhood_group	categorical/text	35796	0	0.00	1
neighbourhood	categorical/text	35796	0	0.00	16
latitude	numeric	35796	0	0.00	3896
longitude	numeric	35796	0	0.00	3938
room_type	categorical/text	35796	0	0.00	4
price	numeric	35796	11318	31.62	918
minimum_nights	numeric	35796	37	0.10	32
number_of_reviews	numeric	35796	0	0.00	813
last_review	categorical/text	35796	3234	9.03	807
reviews_per_month	numeric	35796	3234	9.03	1161
calculated_host_listings_count	numeric	35796	0	0.00	122
availability_365	numeric	35796	0	0.00	366
number_of_reviews_ltm	numeric	35796	0	0.00	165
license	categorical/text	35796	35796	100.00	0
month	categorical/text	35796	0	0.00	11
year	categorical/text	35796	0	0.00	2

Deliverable 4 — Data Cleaning

Cleaned Airbnb Dataset

```
airbnb_cleaned <- read_csv(
  "output_data/Deliverable4_output/Airbnb_listings_cleaned.csv",
  show_col_types = FALSE
)

glimpse(airbnb_cleaned)
```

```
## Rows: 35,796
## Columns: 10
## $ id                <dbl> 4.018377e+07, 1.149729e+18, 1.097487e+06, 1.339648e+1~
## $ neighbourhood     <chr> "Harewood Ward", "Central Ward", "Fendalton Ward", "B~
## $ latitude          <dbl> -43.47002, -43.53270, -43.51149, -43.48286, -43.53844~
## $ longitude         <dbl> 172.5709, 172.6640, 172.6254, 172.6687, 172.5790, 172~
## $ room_type         <chr> "Private room", "Private room", "Private room", "Enti~
## $ price             <dbl> NA, NA, 166, NA, 126, 585, 145, 450, NA, NA, NA, 170,~
## $ availability_365 <dbl> 317, 0, 221, 0, 359, 78, 348, 363, 4, 0, 0, 247, 116,~
## $ month             <chr> "October", "October", "October", "October", "October"~
## $ year              <dbl> 2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025,~
## $ minimum_nights    <dbl> 1, 1, 1, 2, 1, 2, 1, 2, 1, 1, 7, 1, 1, 1, 1, 1, 3, 1,~
```

Example Cleaned Airbnb Data

```
airbnb_cleaned |>
  select(
    id,
    neighbourhood,
```

```
    room_type,
    price,
    availability_365,
    month,
    year
  ) |>
  head(6) |>
  kable(
    caption = "Cleaned Airbnb Data",
    digits = 2
  )
```

Table 3: Cleaned Airbnb Data

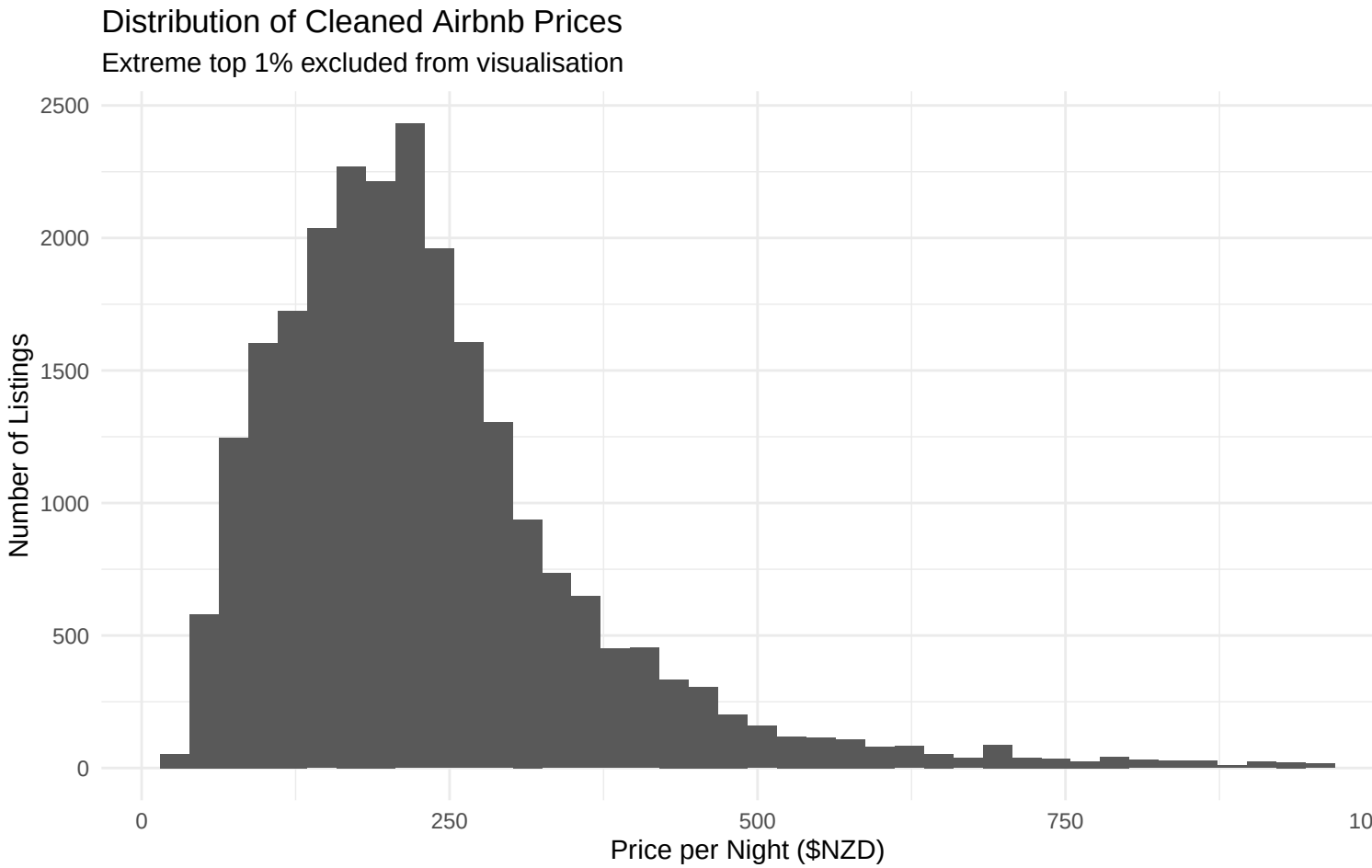
id	neighbourhood	room_type	price	availability_365	month	year
4.018377e+07	Harewood Ward	Private room	NA	317	October	2025
1.149729e+18	Central Ward	Private room	NA	0	October	2025
1.097487e+06	Fendalton Ward	Private room	166	221	October	2025
1.339648e+18	Burwood Ward	Entire home/apt	NA	0	October	2025
1.211070e+18	Riccarton Ward	Private room	126	359	October	2025
1.454661e+18	Fendalton Ward	Entire home/apt	585	78	October	2025

Cleaned Airbnb Price Distribution

The extreme top 1% of prices are excluded from this graph only so the normal price distribution is easier to read. The original data is not changed.

```
price_limit <- quantile(
  airbnb_cleaned$price,
  0.99,
  na.rm = TRUE
)

airbnb_cleaned |>
  filter(
    !is.na(price),
    price <= price_limit
  ) |>
  ggplot(aes(x = price)) +
  geom_histogram(
    bins = 40
  ) +
  labs(
    title = "Distribution of Cleaned Airbnb Prices",
    subtitle = "Extreme top 1% excluded from visualisation",
    x = "Price per Night ($NZD)",
    y = "Number of Listings"
  ) +
  theme_minimal()
```



Cleaned Tenancy Dataset

```
tenancy_cleaned <- read_csv(
  "output_data/Deliverable4_output/tenancy_cleaned.csv",
  show_col_types = FALSE
)

glimpse(tenancy_cleaned)
```

```
## Rows: 26,991
## Columns: 12
## $ TimeFrame                <date> 2026-04-01, 2026-04-01, 2026-04-01, 2026-04~
## $ `Location Id`            <dbl> 100100, 100600, 100700, 100800, 101100, 1028~
## $ `Dwelling Type`          <chr> "ALL", "ALL", "ALL", "ALL", "ALL", "ALL", "A~
## $ `Number Of Beds`         <chr> "1", "1", "1", "1", "1", "1", "1", "1", "1",~
## $ `Total Bonds`            <dbl> 6, 6, 6, 15, 6, 6, 9, 9, 12, 12, 9, 12, 6, 1~
## $ `Active Bonds`           <dbl> 9, 12, 30, 51, 36, 21, 33, 27, 69, 42, 27, 2~
## $ `Closed Bonds`           <dbl> 0, 3, 3, 0, 3, 3, 3, 3, 3, 3, 0, 0, 3, 0, 3,~
## $ `Median Rent`            <dbl> 425, 588, 550, 500, 570, 698, 580, 650, 470,~
## $ `Geometric Mean Rent`    <dbl> 408, 585, 530, 520, 536, 649, 482, 606, 451,~
## $ `Upper Quartile Rent`    <dbl> 475, 720, 579, 550, 621, 739, 590, 710, 550,~
## $ `Lower Quartile Rent`    <dbl> 370, 545, 473, 485, 490, 575, 390, 530, 388,~
## $ `Log Std Dev Weekly Rent` <dbl> 0.2834, 0.2550, 0.1588, 0.0970, 0.1942, 0.29~
```

Example Cleaned Tenancy Data

```
tenancy_cleaned |>
  select(
    TimeFrame,
    `Location Id`,
    `Dwelling Type`,
    `Number Of Beds`,
    `Active Bonds`,
    `Median Rent`
  ) |>
  head(6) |>
  kable(
    caption = "Cleaned Tenancy Data",
    digits = 2
  )
```

Table 4: Cleaned Tenancy Data

TimeFrame	Location Id	Dwelling Type	Number Of Beds	Active Bonds	Median Rent
2026-04-01	100100	ALL	1	9	425
2026-04-01	100600	ALL	1	12	588
2026-04-01	100700	ALL	1	30	550
2026-04-01	100800	ALL	1	51	500
2026-04-01	101100	ALL	1	36	570
2026-04-01	102800	ALL	1	21	698

Deliverable 5 — Data Enrichment and Comparison

Airbnb Listings with SA2 Codes

```
airbnb_sa2 <- read_csv(
  "output_data/Deliverable5_output/Airbnb_listings_sa2.csv",
  show_col_types = FALSE
)

glimpse(airbnb_sa2)
```

```
## Rows: 35,796
## Columns: 11
## $ id               <dbl> 4.018377e+07, 1.149729e+18, 1.097487e+06, 1.339648e+1~
## $ neighbourhood    <chr> "Harewood Ward", "Central Ward", "Fendalton Ward", "B~
## $ latitude         <dbl> -43.47002, -43.53270, -43.51149, -43.48286, -43.53844~
## $ longitude        <dbl> 172.5709, 172.6640, 172.6254, 172.6687, 172.5790, 172~
## $ room_type        <chr> "Private room", "Private room", "Private room", "Enti~
## $ price            <dbl> NA, NA, 166, NA, 126, 585, 145, 450, NA, NA, NA, 170,~
## $ availability_365 <dbl> 317, 0, 221, 0, 359, 78, 348, 363, 4, 0, 0, 247, 116,~
## $ month            <chr> "October", "October", "October", "October", "October"~
## $ year             <dbl> 2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025, 2025,~
## $ minimum_nights   <dbl> 1, 1, 1, 2, 1, 2, 1, 2, 1, 1, 7, 1, 1, 1, 1, 1, 3, 1,~
## $ sa2_code         <dbl> 316700, 327902, 322200, 321302, 322700, 322200, 32520~
```

Example Airbnb Listings with SA2 Codes

```
airbnb_sa2 |>
  select(
    id,
    neighbourhood,
    room_type,
    price,
    month,
    year,
    sa2_code
  ) |>
  head(6) |>
  kable(
    caption = "Airbnb Listings with SA2 Codes",
    digits = 2
  )
```

Table 5: Airbnb Listings with SA2 Codes

id	neighbourhood	room_type	price	month	year	sa2_code
4.018377e+07	Harewood Ward	Private room	NA	October	2025	316700
1.149729e+18	Central Ward	Private room	NA	October	2025	327902
1.097487e+06	Fendalton Ward	Private room	166	October	2025	322200
1.339648e+18	Burwood Ward	Entire home/apt	NA	October	2025	321302
1.211070e+18	Riccarton Ward	Private room	126	October	2025	322700
1.454661e+18	Fendalton Ward	Entire home/apt	585	October	2025	322200

Joined Airbnb and Tenancy Dataset

```
joined_data <- read_csv(
  "output_data/Deliverable5_output/Airbnb_bond_joined_final.csv",
  show_col_types = FALSE
)

glimpse(joined_data)
```

```
## Rows: 35,796
## Columns: 23
## $ id                    <dbl> 4.018377e+07, 1.149729e+18, 1.097487e+06, 1.~
## $ neighbourhood         <chr> "Harewood Ward", "Central Ward", "Fendalton ~
## $ latitude              <dbl> -43.47002, -43.53270, -43.51149, -43.48286, ~
## $ longitude             <dbl> 172.5709, 172.6640, 172.6254, 172.6687, 172.~
## $ room_type             <chr> "Private room", "Private room", "Private roo~
## $ price                 <dbl> NA, NA, 166, NA, 126, 585, 145, 450, NA, NA,~
## $ availability_365      <dbl> 317, 0, 221, 0, 359, 78, 348, 363, 4, 0, 0, ~
## $ month                 <chr> "October", "October", "October", "October", ~
## $ year                  <dbl> 2025, 2025, 2025, 2025, 2025, 2025, 2025, 20~
## $ minimum_nights        <dbl> 1, 1, 1, 2, 1, 2, 1, 2, 1, 1, 7, 1, 1, 1, 1,~
## $ sa2_code              <dbl> 316700, 327902, 322200, 321302, 322700, 3222~
## $ quarter               <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4,~
## $ TimeFrame             <date> NA, NA, 2025-10-01, NA, 2025-10-01, 2025-10~
## $ `Dwelling Type`       <chr> NA, NA, "ALL", NA, "ALL", "ALL", "ALL", "ALL~
```

```
## $ `Number Of Beds`          <chr> NA, NA, "ALL", NA, "ALL", "ALL", "ALL", "ALL~
## $ `Total Bonds`             <dbl> NA, NA, 30, NA, 93, 30, 24, 24, 21, 24, NA, ~
## $ `Active Bonds`            <dbl> NA, NA, 261, NA, 549, 261, 297, 153, 75, 153~
## $ `Closed Bonds`            <dbl> NA, NA, 21, NA, 27, 21, 12, 15, 6, 15, NA, N~
## $ `Median Rent`             <dbl> NA, NA, 723, NA, 780, 723, 555, 710, 680, 71~
## $ `Geometric Mean Rent`     <dbl> NA, NA, 664, NA, 660, 664, 578, 603, 707, 60~
## $ `Upper Quartile Rent`     <dbl> NA, NA, 899, NA, 1165, 899, 648, 790, 700, 7~
## $ `Lower Quartile Rent`     <dbl> NA, NA, 514, NA, 471, 514, 464, 444, 670, 44~
## $ `Log Std Dev Weekly Rent` <dbl> NA, NA, 0.4896, NA, 0.7065, 0.4896, 0.2534, ~
```

Example Joined Data

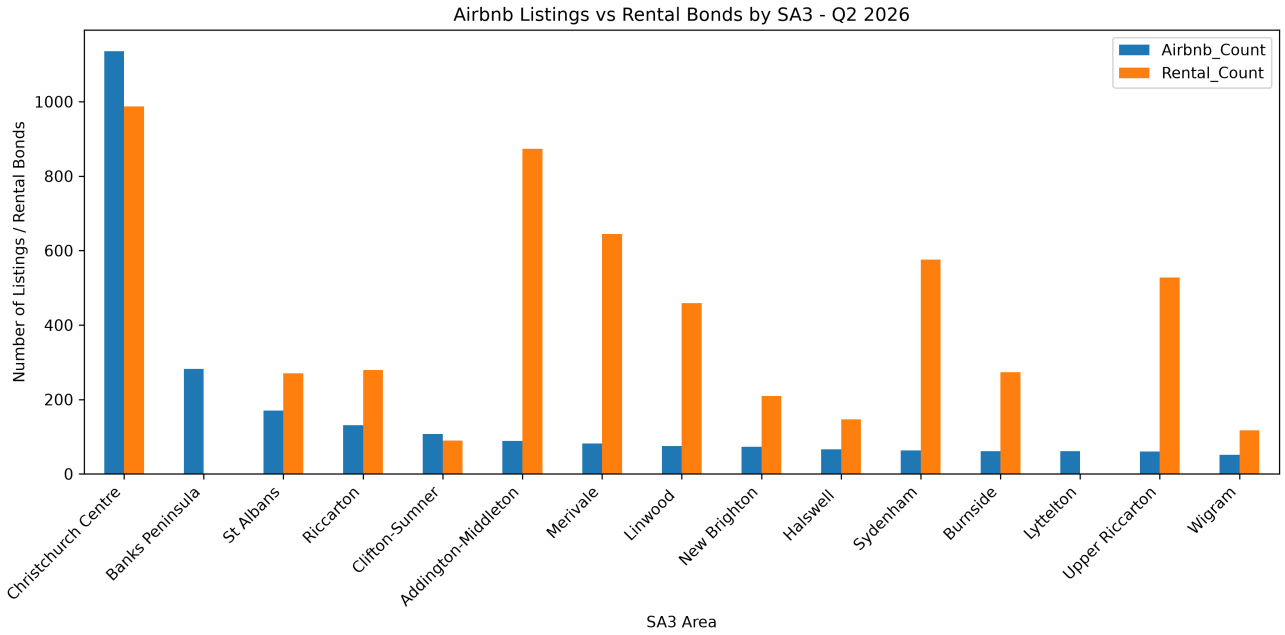
Only the main Airbnb and tenancy fields are shown below so the joined table remains readable.

```
joined_data |>
  select(
    id,
    neighbourhood,
    price,
    month,
    year,
    sa2_code,
    `Active Bonds`,
    `Median Rent`
  ) |>
  head(6) |>
  kable(
    caption = "Joined Airbnb and Tenancy Data",
    digits = 2
  )
```

Table 6: Joined Airbnb and Tenancy Data

id	neighbourhood	price	month	year	sa2_code	Active Bonds	Median Rent
4.018377e+07	Harewood Ward	NA	October	2025	316700	NA	NA
1.149729e+18	Central Ward	NA	October	2025	327902	NA	NA
1.097487e+06	Fendalton Ward	166	October	2025	322200	261	723
1.339648e+18	Burwood Ward	NA	October	2025	321302	NA	NA
1.211070e+18	Riccarton Ward	126	October	2025	322700	549	780
1.454661e+18	Fendalton Ward	585	October	2025	322200	261	723

Airbnb vs Long-Term Rentals by SA3 Area



Christchurch Central Airbnb Prices

```
central_prices <- read_csv(
  "output_data/Deliverable5_output/Christchurch_Central_Airbnb_prices.csv",
  show_col_types = FALSE
)

glimpse(central_prices)
```

```
## Rows: 160
## Columns: 3
## $ neighbourhood <chr> "Central Ward", "Central Ward", "Central Ward", "Central~
## $ sa2_code      <dbl> 326600, 326600, 326600, 326600, 326600, 326600, 326600, ~
## $ price         <dbl> 140, 319, 43654, NA, NA, NA, 222, 501, NA, NA, 233, NA, ~
```

```
central_prices |>
  select(
    neighbourhood,
    sa2_code,
    price
  ) |>
  head(10) |>
  kable(
    caption = "Christchurch Central Airbnb Prices",
    digits = 2
  )
```

Table 7: Christchurch Central Airbnb Prices

neighbourhood	sa2_code	price
Central Ward	326600	140
Central Ward	326600	319
Central Ward	326600	43654
Central Ward	326600	NA
Central Ward	326600	NA
Central Ward	326600	NA
Central Ward	326600	222
Central Ward	326600	501
Central Ward	326600	NA
Central Ward	326600	NA

Largest Short-Term and Long-Term Rental Price Differences

```
rent_difference <- read_csv(
  "output_data/Deliverable5_output/listings_sa3_rent_diff.csv",
  show_col_types = FALSE
)

glimpse(rent_difference)
```

```
## Rows: 13,858
## Columns: 25
## $ id                        <dbl> 1.097487e+06, 1.211070e+18, 1.454661e+18, 5.~
## $ neighbourhood             <chr> "Fendalton Ward", "Riccarton Ward", "Fendalt~
## $ latitude                  <dbl> -43.51149, -43.53844, -43.51108, -43.52902, ~
## $ longitude                 <dbl> 172.6254, 172.5790, 172.6239, 172.6107, 172.~
## $ room_type                 <chr> "Private room", "Private room", "Entire home~
## $ price                     <dbl> 166, 126, 585, 145, 450, 110, 148, 141, 190,~
## $ availability_365          <dbl> 221, 359, 78, 348, 363, 116, 132, 311, 282, ~
## $ month                     <chr> "October", "October", "October", "October", ~
## $ year                      <dbl> 2025, 2025, 2025, 2025, 2025, 2025, 2025, 20~
## $ minimum_nights            <dbl> 1, 1, 2, 1, 2, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1,~
## $ sa2_code                  <dbl> 322200, 322700, 322200, 325200, 332000, 3270~
## $ quarter                   <dbl> 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4, 4,~
## $ TimeFrame                 <date> 2025-10-01, 2025-10-01, 2025-10-01, 2025-10~
## $ `Dwelling Type`           <chr> "ALL", "ALL", "ALL", "ALL", "ALL", "ALL", "A~
## $ `Number Of Beds`          <chr> "ALL", "ALL", "ALL", "ALL", "ALL", "ALL", "A~
## $ `Total Bonds`             <dbl> 30, 93, 30, 24, 24, 144, 51, 81, 99, 39, 45,~
## $ `Active Bonds`            <dbl> 261, 549, 261, 297, 153, 954, 570, 1107, 732~
## $ `Closed Bonds`            <dbl> 21, 27, 21, 12, 15, 69, 24, 48, 42, 21, 21, ~
## $ `Median Rent`             <dbl> 723, 780, 723, 555, 710, 499, 530, 495, 850,~
## $ `Geometric Mean Rent`     <dbl> 664, 660, 664, 578, 603, 455, 485, 448, 774,~
## $ `Upper Quartile Rent`     <dbl> 899, 1165, 899, 648, 790, 553, 580, 550, 121~
## $ `Lower Quartile Rent`     <dbl> 514, 471, 514, 464, 444, 408, 475, 420, 560,~
## $ `Log Std Dev Weekly Rent` <dbl> 0.4896, 0.7065, 0.4896, 0.2534, 0.4896, 0.37~
## $ `price difference`        <dbl> 498, 222, 3431, 437, 2547, 315, 551, 539, 55~
## $ SA32023_code              <dbl> 60820, 60920, 60820, 60910, 61290, 60900, 61~
```

Example Rental Price Differences

```
rent_difference |>
  select(
    neighbourhood,
    price,
    sa2_code,
    `Median Rent`,
    `price difference`,
    SA32023_code
  ) |>
  head(10) |>
  kable(
    caption = "Short-Term vs Long-Term Rental Price Differences",
    digits = 2
  )
```

Table 8: Short-Term vs Long-Term Rental Price Differences

neighbourhood	price	sa2_code	Median Rent	price difference	SA32023_code
Fendalton Ward	166	322200	723	498	60820
Riccarton Ward	126	322700	780	222	60920
Fendalton Ward	585	322200	723	3431	60820
Riccarton Ward	145	325200	555	437	60910
Heathcote Ward	450	332000	710	2547	61290
Central Ward	110	327000	499	315	60900
Heathcote Ward	148	329400	530	551	61120
Linwood Ward	141	328900	495	539	61230
Riccarton Ward	190	321800	850	556	60920
Central Ward	174	327100	500	698	60900

Pipeline Summary

The automated pipeline performs the following stages:

1. Combines the monthly Airbnb datasets.
2. Produces the Deliverable 3 analyses.
3. Cleans the Airbnb and tenancy datasets.
4. Assigns geographic SA2 codes to Airbnb listings.
5. Joins Airbnb and tenancy information.
6. Performs the Deliverable 5 analyses.
7. Generates this report automatically.

The full pipeline can be executed using:

```
make all
```